

1. Samal is using a computer software to design the layout of his new game room. The base of the new game room creates $GAMR$ with coordinates $G(-7, -5)$, $A(3, -9)$, $M(5, -4)$, and $R(-5, 0)$.

- a. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{GA}$	
$\overline{AM}$	
$\overline{MR}$	
$\overline{GR}$	

- b. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{GM}$	
$\overline{AR}$	

- c. Finish the statement.

$GAMR$ is classified as a _____ because ...

- d. If 1 unit on the coordinate plane equals 4 feet (ft), what are the dimensions of the base of the new game room? Round to the nearest thousandth.

2. $STLE$ has vertices at $S(-5,6)$, $T(0,-3)$, $L(6,-5)$, and $E(1,4)$.

$STLE$ is classified as a(n) _____ because ...

3. Franco is designing a template for his new goat pen. He created two quadrilaterals using a computer software program.

$MFCR$	$AMBT$
$M(-1,5)$, $F(-9,-3)$, $C(-5,-7)$, $R(3,1)$	$A(-2,-3)$, $M(4,5)$, $B(0,7)$, $T(-6,0)$

- a. Which quadrilateral can be classified as a rectangle? Show your work.
- b. List the properties of a rectangle that you used.
- c. If 1 unit of the coordinate plane equals 7 feet (ft), what are the measurements of the sides of the rectangle?
4. Akilah was graphing square $STVE$, but mixed up her vertices. She knows that point S has coordinates $(0,4)$ and point E has coordinates $(8,0)$. Her two options for points T and V are shown.

Point T	Point V
$(-4,-4)$	$(-2,-8)$
$(4,-4)$	$(4,-8)$

- a. Which coordinates are correct for points T and V ? Justify your answer or show your work.
- b. Which properties of a square did you use to determine the correct coordinates for points T and V ?